THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The exponent of 2 in the prime factorisation of 2400 is: 1

(a) 7 **(b)** 6 **(c)** 8 **(d)** 5

2. The exponent of 2 in the prime factorisation of 4320 is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 8

3. If HCF(144, 198) = 18, then LCM(144, 198) is: 1

(a) 28512 **(b)** 342 **(c)** 2592 **(d)** 1584

4. The product of a non-zero rational and an irrational number such as $4\sqrt{7}$ is: 1

(a) sometimes rational **(b)** always rational

(c) always an integer **(d)** always irrational

5. The decimal expansion of $\frac{15}{1600}$ terminates after how many places of decimals? 1

(a) 9 **(b)** 6 **(c)** 7 **(d)** 8

6. Which of the following is irrational? 1

(a) $\sqrt{121} - 5$ **(b)** $5 \times \sqrt{25}$ **(c)** $\sqrt{25} + 5$ **(d)** $\sqrt{11} + 5$

7. For which of the following is n such that 21^n ends with the digit 0? 1

(a) every n **(b)** no natural number n

(c) $n = 2$ only **(d)** only even n

8. The exponent of 2 in the prime factorisation of 3584 is: 1

(a) 9 **(b)** 12 **(c)** 10 **(d)** 11

9. The exponent of 3 in the prime factorisation of 2646 is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

10. If HCF(95, 152) = 19, then LCM(95, 152) is: 1

(a) 247 **(b)** 14440 **(c)** 760 **(d)** 1805

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why $5 \times 9 + 5$ is a composite number. 2

12. Check whether 35^n can end with the digit 0 for any natural number n. 2

13. Given that HCF(144,198)=18, find LCM(144,198). 2

14. Explain why $3 \times 8 + 3$ is a composite number. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{11}$ is irrational. 3

16. Show that $4 + 5\sqrt{11}$ is irrational, given that $\sqrt{11}$ is irrational. 3

17. Find the HCF and LCM of 306 and 657 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

OR

Prove that $\sqrt{13}$ is irrational. 3

18. Show that $6 + 2\sqrt{13}$ is irrational, given that $\sqrt{13}$ is irrational. 3

OR

Find the HCF and LCM of 95 and 152 by prime factorisation and verify HCF $\times$ LCM $=$ product. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that $\sqrt{31}$ is irrational. 5

OR

Prove that $\sqrt{5}$ is irrational and hence that $3 - 4\sqrt{5}$ is irrational. 5

20. Prove that $\sqrt{37}$ is irrational. 5

OR

Prove that $\sqrt{7}$ is irrational and hence that $2 - 3\sqrt{7}$ is irrational. 5

END OF QUESTION PAPER — Total: 40 Marks

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